SHREYASH MESHRAM

Full-Stack & AI Engineer

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PERSONAL PROFILE

Full-Stack & AI Engineer who has recently completed an MSc in Business Analytics at the University of Nottingham, building on a BE in Computer Engineering. Skilled in building Python back ends (FastAPI, Django), React interfaces and applied machine-learning pipelines – from privacy-first LLM tools to computer-vision and NLP systems – with two software development internships and three published papers. Seeking graduate or junior roles in full-stack engineering, AI engineering or applied machine learning.

EDUCATION

MSc Business Analytics – University of Nottingham, UK — Sep 2025 – Sep 2026

- **Dissertation:** 'Identifying feature drift using variable importance and MCR' – featured under Selected Projects.
- **Coursework projects:** retail customer segmentation (five segments from 3,000 loyalty-card customers, using K-Means and PCA); store KPI analysis of £1.29M of transactions (SQL, Databricks, Tableau); Bluesky brand intelligence (LDA topic modelling, keyword networks); calibrated business-risk prediction pipeline.
- **Key modules:** Machine Learning and Predictive Analytics; Data at Scale: Management, Processing and Visualisation; Leading Big Data Business Projects; Advanced Operations Analysis; Supply Chain Planning and Management.

BE Computer Engineering – Sandip Institute of Technology & Research Centre, Nashik, India — Jun 2022 – Jun 2025

- **Key modules:** Machine Learning, Deep Learning, Artificial Intelligence, Data Science and Big Data Analytics, Cloud Computing, Software Engineering, Data Structures and Algorithms.

Diploma in Computer Engineering – Sandip Polytechnic, Nashik, India — Jun 2019 – Jun 2022

TECHNICAL EXPERIENCE

Python Developer Intern – Arohi Softwares — Aug 2024 – Feb 2025

- Built responsive Django interfaces from UI/UX designs, achieving a 95% satisfaction rate in user acceptance testing.
- Cut API p50 latency by ~40% through query batching and indexed database lookups.
- Integrated REST APIs and owned relational schema design in PostgreSQL, maintaining data integrity across features.
- Shipped six feature releases on a weekly cadence with zero rollbacks as part of a cross-functional Agile team.

Software Developer Intern – NerdTech — Mar 2024 – Apr 2024

- Developed client software with a focus on user-centred design and robust back-end systems.
- Reduced runtime errors by 30% by introducing an end-to-end test harness, and closed 22 backlog bugs in six weeks.

Machine Learning with AI (Python) Trainee – Knowledge Solutions India — Dec 2021 – Feb 2022

- Built a Heart Disease Predictor ML pipeline in a team using scikit-learn and Pandas (89% recall, 5-fold CV).
- Developed a demand-forecasting model that reduced a partner's inventory costs by 15%.

ADDITIONAL EXPERIENCE

Kitchen Porter (Part-Time) – Panda Mami, Nottingham — Jun 2026 – Present

- Keep kitchen equipment, crockery and work areas clean and service-ready during busy buffet services.
- Follow food hygiene and COSHH procedures, and help chefs with deliveries, stock rotation and basic prep.
- Work in a fast-paced kitchen team at peak times, building reliability, teamwork and time management.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, C, C++
- **Back end & web:** FastAPI, Django, Flask, REST APIs, React, React Native, Tailwind CSS
- **AI & ML:** scikit-learn, PyTorch, TensorFlow, Hugging Face (BERT, FinBERT), LangChain, Ollama, OpenCV, MediaPipe, SHAP
- **Data & analytics:** Pandas, NumPy, Databricks, Tableau, Matplotlib, NLTK, Gensim, NetworkX
- **Databases & cloud:** PostgreSQL (pgvector), MySQL, MariaDB, MongoDB, Firebase, AWS, Google Cloud, Azure
- **Tools & practices:** Git, Docker, CI/CD, Vercel, pytest, Agile, Figma, Adobe Photoshop, Premiere Pro, After Effects

SELECTED PROJECTS

Identifying feature drift using variable importance and MCR — **MSc dissertation · 2026**
Python · scikit-learn (random forests) · SHAP · River (DDM, ADWIN) · Pandas · Matplotlib — GitHub · Case study

- Tested when variable-importance monitoring can warn of concept drift before error-based detectors, using a 200,000-instance rotating-hyperplane benchmark and a Boolean harness with exactly known ground truth.
- Showed that refitted SHAP importance alarmed seven windows (~7,000 instances) before accuracy degraded – matching DDM and well ahead of ADWIN – while frozen-model SHAP and random-forest vote dominance stayed silent, as theory predicts for conditional drift.
- Calibrated every signal with bootstrap null bands and permutation negative controls: once the drift was destroyed, the refitted alarm moved from window 1 to window 106.

Accident Black Spot Detection – road-safety system (published) — BE major project
Android · Firebase · Google Maps API · ECLAT | 2026 rebuild: React Native (Expo) · FastAPI · DBSCAN — GitHub

- Built an Android app with Firebase and the Google Maps API that alerts users to nearby accident-prone zones in real time, mining historical accident data with ECLAT and adding SOS alerts for emergencies.
- Rebuilt it in 2026 as a full-stack system: a React Native app, a web moderation dashboard so only a human can publish a warning, and a FastAPI service using DBSCAN and a hand-written ECLAT miner (1,500+ tests).

FinCast AI – stock market analysis platform — BE major project (alternative)
FastAPI · React · scikit-learn · FinBERT · Pandas · MySQL — GitHub · Case study

- Built and deployed a full-stack app for NSE-listed stocks in which an RBF-kernel SVR forecasts seven trading days ahead, FinBERT news sentiment adjusts the forecast, and sentiment plus RSI drive a Buy/Sell/Hold/Avoid call.
- Charted EMA, MACD, Bollinger Bands and OBV in a React front end, backed by a FastAPI prediction service and a separate MySQL-backed account service with bcrypt-hashed passwords.

Sign Language Recognition – real-time gesture-to-speech (published) — Diploma major project
Python · TensorFlow · MediaPipe · OpenCV · gTTS — GitHub · Case study

- Built a real-time recogniser in which MediaPipe extracts 21 hand landmarks, a TensorFlow model classifies the sign and gTTS speaks it aloud; normalising landmarks made recognition independent of hand position in the frame.

ZenSpend – privacy-first AI expense tracker — Personal project
FastAPI · LangChain · PostgreSQL + pgvector · Ollama (Llama 3) · React · Tailwind CSS — GitHub · Case study

- Built a local-first expense tracker whose categorisation, semantic search and chat coach all run on a local Llama 3 via Ollama, so financial data never leaves the machine.
- Paired pgvector similarity search with SQL aggregates in hybrid queries, and eliminated hallucinated totals by routing all arithmetic through SQL instead of the LLM.

FENgine – chess-board OCR — Personal project
FastAPI · OpenCV · PyTorch (CNN) · React · python-chess — GitHub · Case study

- Built a pipeline that converts a single photo of a chess board into FEN and replayable PGN, using OpenCV board detection and perspective correction and a CNN that classifies each of the 64 squares into 13 classes.
- Reached ~97% piece-classification validation accuracy, training largely on synthetic, augmented board renders, with a review step that turns misreads into one-click fixes.

PUBLICATIONS & CERTIFICATIONS

- Gaikwad, G., **Meshram, S.**, Jangale, M., Mahajan, D. and Nevkar, K. (2022) 'Sign Language Recognition using Machine Learning', *International Journal of Scientific Development and Research (IJSDR)*, 7(5), pp. 485–488.
- 'Black Spot Detection Using Android App', *OAIJSE*, paper ref. 978_OAIJSE0545S2479_8899.
- 'Black Spot Detection using Android', *IJASRET*, paper ref. IJASRET0855RV46.
- AWS Knowledge: Cloud Essentials – Training Badge, Amazon Web Services Training and Certification (verified on Credly).